

SymptoSense AI: A Multi-Agent Health Symptom Analytics, Risk Prediction, and Care Routing Ecosystem

SOA PROGRAMMING AND MICROSERVICES Project Review-1

Thursday, August 20th, 2026



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Problem Statement

Emergency departments worldwide are overwhelmed with patient volume, and a lot of the people showing up don't actually need emergency care — they just didn't have anywhere else to turn for advice. The apps that are supposed to help with this are pretty basic: they ask fixed yes/no questions, can't handle it when someone describes several confusing symptoms at once, and never explain why they're suggesting what they're suggesting. So people don't really trust them, and doctors can't rely on them either. On top of that, people in remote or underserved areas often don't have quick access to a doctor at all, so they either wait too long or end up going straight to the ER for things that didn't need it. What's missing is a system that can actually understand what someone is describing in their own words, figure out how serious it might be, and then point them to the right kind of help — whether that's the ER, a video doctor visit, or just home care advice.

Objectives

- To develop a Symptom Analysis Agent that predicts likely associated symptoms and infection type from a partial set of user-reported symptoms using a multi-label classification model.
- To build a risk-tier mapping system that assigns each predicted disease a risk level (High, Moderate, Low) based on a predefined lookup table.
- To implement a Care Routing Agent that applies rule-based logic to convert the risk level into an actionable recommendation (visit ER immediately, book an appointment, or self-care guidance).
- To integrate all three agents into a service-oriented architecture using Spring Boot microservices, API Gateway, Eureka service discovery, and Kafka-based event communication.

Literature Survey

Authors	Title of the Paper	Methodology	Inferences	Limitation/Future work
Ibrahim et al., 2026	TriAgent: Multi-Agent Crisis Clinical Decision Support	Multi-agent LLM orchestration	High-accuracy, explainable triage	Needs prospective clinical validation
Iqbal et al., 2016	Application of Intelligent Agents in Healthcare: Review	Literature review	MAS models complex healthcare systems well	Security and interoperability gaps
Gadekallu et al.	Agentic AI in Healthcare 5.0: Reference Architecture	Survey + reference architecture	Agentic AI enables human-centric care	Safety, ethics, and regulation unresolved

Research Gap Identification

Existing systems have several limitations:

- Most symptom-checker apps rely on static keyword matching or rigid decision trees, not learned models.
- Existing ML-based triage tools predict a condition but rarely explain why, leaving clinicians and patients without reasoning they can trust.
- Risk-scoring models mostly output a single probability, without independent safety validation for high-risk cases.
- Existing systems struggle to reliably distinguish genuinely urgent cases from common, low-risk symptom overlaps.

Innovation, Creativity, and Novelty

Proposed Innovations

- Multi-agent pipeline connecting symptom prediction, disease-risk classification, and automated care routing.
- Classifies specific diseases from symptom patterns, not just generic categories.
- Risk-tier mapping (High/Moderate/Low) applied automatically to each predicted disease.
- Fully rule-based care routing — no manual decision needed, instant actionable guidance.

Novel Features

Symptom expansion prediction (partial → full symptom profile)

Disease-specific risk classification

Automated risk-to-action mapping

Instant care guidance (ER / appointment / self-care)

Feasibility and Project

Project Timeline

